Nick DiSanto

https://nickdisanto.github.io nicolas.c.disanto@vanderbilt.edu

Education

Vanderbilt University Jan. 2025 – Present
PhD in Computer Science
GPA: 3.94/4.0. Advised by Dr. Ipek Oguz.

California Baptist University Aug. 2019 – Apr. 2023
B.S. in Computer Science, *summa cum laude*
GPA: 4.0/4.0. Ranked #1 in graduating class.

Research Experience

Vanderbilt Medical Image Computing Lab (MedICL) Jan. 2025 – Present
Graduate Student Researcher
- Developing deep learning methods for image restoration and generation, with emphasis on 3D medical image volumes
- Designing registration and volumetric inpainting models for high-resolution 3D OCT/OCTA data to correct motion-induced slice corruption
- Exploring how learned representations can maintain robustness across diverse imaging tasks

CalBaptist ML & NLP Lab Jan. 2022 – Dec. 2023
Lead Undergraduate Researcher
- Developed representation learning models to identify patterns in large-scale online data
- Proposed a novel data preprocessing approach to help generalize medical imaging models
- Led end-to-end research efforts across three papers, from inception through execution

Industry Experience

GE Vernova Dec. 2023 – Jan. 2025
Software Engineer
- Led a wildfire forecasting initiative, including the end-to-end model training, visualization, and productization of ML pipelines for power-grid applications
- Presented results of wildfire risk forecasting analytic to the C-Suite, contributing to secured funding for continued integration
- Recognized as a 2024 GE Vernova Changemaker for co-leading a company-wide initiative to integrate GenAI into Electrification Software products

General Electric Jun – Aug 2022; Jun – Aug 2023
Software Engineering Intern
- Led a database optimization initiative resulting in annual cost savings of several thousand dollars
- Developed and optimized microservices supporting large-scale data ingestion pipelines

Keysight Technologies Jun. 2021 – Aug. 2021
Software Engineering Intern
- Developed a web platform enabling clients to customize product configurations
- Performed data analysis that eliminated 40% of redundant product-option configurations

Publications

VAMOS-OCTA: Vessel-Aware Multi-Axis Orthogonal Supervision for Inpainting Motion-Corrupted OCT Angiography Volumes
N DiSanto, EK Aghdam, H Liu, JJ Watson, YK Tao, H Li, I Oguz.
SPIE Medical Imaging, 2026 **[Oral Deep-Dive]** **[Best Student Paper]**. [pdf] [code]

Monocular Absolute Depth Estimation from Endoscopy via Domain-Invariant Feature Learning and Latent Consistency
H Li, D Lu, J d'Almeida, D Isik, EK Aghdam, **N DiSanto**, A Acar, S Sharma, JY Wu, RJ Webster III, Ipek Oguz.
SPIE Medical Imaging, 2026 **[Oral]**. [pdf] [code]

From Geometry to Intensity: A Coarse-to-Fine Pipeline for Unsupervised 3D Ultrasound Stitching
X Yao, R Yu, D Lu, **N DiSanto**, EK Aghdam, KJ Oguine, G Arenas, B Oguz, A Pouch, N Schwartz, B Byram, I Oguz.
SPIE Medical Imaging, 2026.

A Benchmark Study of Methods for Surgical Instrument Segmentation in Central Airway Endoscopy
D Isik, KJ Oguine, **N DiSanto**, J d'Almeida, RJ Webster III, I Oguz, H Li.
SPIE Medical Imaging, 2026 **[Best Poster: Honorable Mention]**.

LOTUS: Latent Outpainting Diffusion Model for Three-Dimensional Ultrasound Stitching
X Yao, R Yu, **N DiSanto**, E Aghdam, KJ Oguine, D Lu, A Lou, J Wang, D Hu, G Arenas, B Oguz, A Pouch, N Schwartz, B Byram, I Oguz.
Medical Imaging with Deep Learning (MIDL), 2025 **[Oral]**. [pdf] [code]

Spatial Analysis of Social Media's Proxies for Human Emotion and Cognition
A Corso, **N DiSanto**, N Corso, E Lee.
International Conference on Information, 2024.

Teaching Experience

EGR121: **Intro to C++**, *Teaching Assistant*	SP 2023
EGR225: **Discrete Structures**, *Teaching Assistant*	FA 2022
CSC312: **Algorithms**, *Tutor*	SP/FA 2022
EGR329: **Computer Architecture**, *Tutor*	SP/FA 2022
PHY201: **Physics for Engineers**, *Teaching Assistant*	SP/FA 2021

Awards & Honors

2024 GE Vernova Changemaker (selected out of 1200 nominees)	Nov. 2024
CS Outstanding Student Award (ranked #1/40)	Apr. 2023
Inducted into Alpha Chi Honor Society (top 10%)	Apr. 2022
Undergraduate Physics Performance Award	Dec. 2021
President's List (every semester of undergraduate study)	FA 2019 – SP 2023

Professional Memberships

Institute of Electrical and Electronics Engineers (IEEE)	Jan. 2025 – Present
Society of Photo-Optical Instrumentation Engineers (SPIE)	Jul. 2025 – Present